ASM-FreqNet: Adaptive Soft Frequency Masking for Generalizable Deepfake Image Detection

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Abstract. Recent advances in generative models have enabled the synthesis of highly realistic images, posing significant challenges to digital media authentication. Frequency-domain detection methods have shown strong generalization by exploiting spectral artifacts introduced during image generation. This paper proposes ASM-FreqNet, an enhanced FreqNet architecture based on an Adaptive Region-based Soft Masking (ASM) mechanism. Instead of suppressing low-frequency components with fixed zero values, ASM introduces learnable sigmoid-constrained weights that adaptively preserve global structural information while retaining high-frequency artifacts. The proposed mechanism is integrated into both the Mask-Softened High-Frequency Representation of Images (MS-HFRI) and Mask-Softened High-Frequency Representation of Features (MS-HFRF) modules, enabling joint exploitation of structural and spectral cues. Experiments conducted under identical training settings on a 50,000-image subset of the AI-Generated-vs-Real dataset demonstrate that ASM-FreqNet achieves 74.03% accuracy, outperforming the original FreqNet by 1.87 percentage points while maintaining the same lightweight architecture with approximately 1.85 million parameters. These results demonstrate that adaptive preservation of low-frequency information effectively complements high-frequency artifacts, improving the generalization capability of frequency-domain deepfake image detection without increasing model complexity.

Keywords: Deepfake Image Detection, FreqNet, Frequency-Domain Detection, Adaptive Soft Masking, AI-Generated Images, Spectral Artifacts